



The analysis process

The 4 phases of analysis

1. Organize data
2. Format and adjust data
3. Get input from others
4. Transform data

Welcome back. It's great to see you again. So let's talk about analysis. We've learned how to ask the right questions, prepare data for exploration, and then process that data to make sure it's squeaky clean. Now it's time for the heart of the process: the actual analysis! Finally, right? But what is analysis?

Basically, analysis is the process used to make sense of the data collected. It means taking the right steps to proceed and think about your data in different ways. The goal of analysis is to identify trends and relationships within the data so that you can accurately answer the question you're asking. To do this, you should stick to the 4 phases of

analysis: **organize data**, **format and adjust data**, **get input from others**, and **transform data** by observing relationships between data points and making calculations. Let's apply the 4 phases of analysis to a real-world scenario. Imagine you want to buy a gift for your friend Zara's wedding. The problem is you're not sure what to get her.

Fortunately, you have a ton of data from her wedding website. But instead of reading all the data on her website and scrolling through a photo album of her and her partner, you go straight to the online registry, a wish list of gifts they'd enjoy. The registry is like a dataset that you can analyze to make a decision. **Now that you're checking out organized data in the registry**, you want to make sure that the list of data, or gifts in this case, is formatted in a way that's easy to reference. **Formatting data streamlines things and saves you time.** Scrolling through hundreds of gifts can be time-consuming.

Instead, you can **adjust the data in a way that makes it easy to digest by filtering and sorting your data.**

You have a budget you want to stick to, so you sort the gift prices from low to high. You then filter prices to include gifts that are within your budget of \$60. You're working with a newly formatted list of data. At this point, **it's good to remember that input from other people can also be really helpful when analyzing information and making decisions.** You can check the list of gifts to figure out if anyone else has already bought any of the items. You realize a few of the items in the list have been purchased, and this informs your decision. When analyzing data, **gaining input from others is important because it gives you a viewpoint you might not understand or have access to.**

On top of gaining input from other people, it's also important to seek out others' **perspectives** early. That way, if they predict any obstacles or challenges, you'll know

beforehand. The people you'll look to for input don't have to be experts to be helpful.

Sometimes all you need is for someone who's familiar with a topic or data you're considering. In our example, that would be Zara's wedding guests who are purchasing gifts from the same online registry. They probably aren't wedding gift experts, but their collaborative effort to mark off the item they purchase can help you figure out what not to buy, which will prevent Zara from getting the same gift twice. In the end, getting input is valuable to your analysis.

This brings us to the last step of the analysis: transforming data. **Transforming data means identifying relationships and patterns between the data, and making calculations based on the data you have.** Going back to our example, you were able to find a gift that you knew Zara would like, and one that fits your budget. You were also able to choose a gift that wasn't already purchased by someone else. By finding the relationship between these data points, you chose, purchased, and sent a gift that would answer the problem you wanted to solve. The beauty of the analysis process is that you probably already analyze situations in your everyday life. Whether you're analyzing data in your personal life or in your career, these four tasks can help you make better decisions.

The more you do it, the more comfortable you'll feel with the process. I hope this gives you a better understanding of the basics of analysis. As we move forward, we'll check out how to locate data for analysis, both in a spreadsheet and using SQL. When you're ready, you can go ahead. See you soon!